

Feiyu Lu, PhD

RESEARCH SCIENTIST

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About Me

Feiyu Lu is a Research Scientist in the Immersive Technology & Spatial Computing Team, Global Technology Applied Research Group at JPMorgan-Chase. He obtained his Ph.D. degree at Virginia Tech advised by Prof. Doug A. Bowman. His research interest lies in the intersections of **Human-AI Interaction**, **Spatial/Multimodal Interaction**, **Artificial Intelligence / Machine Learning**, and **Augmented/Virtual/Mixed Reality**.

Education

Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA, USA

Ph.D. in Computer Science & Applications

Aug 2018 - May 2023

- Research Group: **3D Interaction Group**
- Advisor: **Prof. Doug A. Bowman**
- Dissertation: 📄 **Glanceable AR - Towards a Pervasive and Always-On Augmented Reality Future**
- Committee Members: Prof. Joseph L. Gabbard, Prof. Wallace Lages, Prof. Sang Won Lee, Prof. Yalong Yang, Prof. Steven K. Feiner

Xi'an Jiaotong-Liverpool University (XJTLU)

Suzhou, Jiangsu, China

B.Eng in Computer Engineering

Sep 2014 - Jun 2018

- Research Group: X-CHI Group
- Advisor: **Prof. Hai-Ning Liang**
- Graduated with First-Class Honor (Top 5%)

Research Experience

Immersive Technology & Spatial Computing Team, JPMorganChase

New York, NY, USA

Research Scientist

May 2023 - Current

- I am working as an Applied Research Scientist in the Global Technology Applied Research Group at JPMC.
- I investigate novel integrations of immersive technology, interactive wall displays, and AI/ML techniques such as reinforcement learning, large language models, and large multimodal models to addressing concrete business problems and advancing our clients & employee experiences.
- I work with internal stakeholders, build novel intelligent systems, conduct user-centric research, and deliver tech-transfer-ready proof-of-concept systems to boost employee productivity and client engagement.
- I have contributed to six patents, five full papers, and three extended abstracts at top conferences.
- Intern Mentored: **Dr. Siyou Pei**
- Manager: **Prof. Blair MacIntyre**

Reality Labs Research, Meta Inc.

Redmond, WA, USA

Research Scientist Intern

May 2022 - Aug 2022

- I worked with teams at Reality Labs Research to perform user experience research on AR/VR interactions.
- I researched how user and system factors may influence the perceived human-AI interaction experience in AR/VR.
- I conducted a remote VR study with over 80 participants.
- The work was published and presented at ISMAR'23.
- Mentors: Yan Xu, Brennan Jones, Missie Smith, Laird Malamed & Sophie Kim

Reality Labs Research, Meta Inc.

Remote

Research Intern

May 2021 - Aug 2021

- I worked with teams at Reality Labs Research to perform usability evaluations on intelligent AR interfaces.
- I conducted workshops with UX researchers and designers to identify pain-points during AR interactions.
- I conducted a remote VR study with over 40 participants.
- The work was patented and published at CHI'22.
- Mentors: Yan Xu, Missie Smith & Sophie Kim

3D Interaction Group

Blacksburg, VA, USA

Graduate Researcher

Aug 2018 - May 2023

- I worked as a GRA in the 3D Interaction Group directed by Prof. Doug A. Bowman.
- I worked on exploring interaction techniques for lightweight activation of everyday AR content.
- I worked on designing, implementing and evaluating general-purpose systems for AR HWDs.
- I worked on developing efficient and unobtrusive information access strategies for AR HWDs.
- I worked on evaluating gaze direction visualization techniques in wide-area collaborative AR environments.
- Advisor: Prof. Doug A. Bowman

Selected Publications

I published the majority of my work in premiere conferences / journals on spatial user interactions / human-computer interactions / immersive technologies, including IEEEVR, ISMAR, TVCG and ACM CHI, SUI. The typical acceptance rates of these venues are around 25%. Below is

a selected list of my publications. The full paper list is attached to the end of this resume.

JOURNAL ARTICLES

Revisiting Performance Models of Distal Pointing Tasks in Virtual Reality

Logan Lane, **Feiyu Lu**, Shakiba Davari, Robert J. Teather, Doug A. Bowman

IEEE Transactions on Visualization and Computer Graphics. 2025

DOI: 10.1109/TVCG.2025.3567078

“Where Did My Apps Go?” Supporting Scalable and Transition-Aware Access to Everyday Applications in Head-Worn Augmented Reality

Lu, Feiyu, Leonardo Pavanatto, Shakiba Davari, Lei Zhang, Lee Lisle, Doug A. Bowman

IEEE Transactions on Visualization and Computer Graphics. 2025

DOI: 10.1109/TVCG.2024.3493115

Multiple Monitors or Single Canvas? Evaluating Window Management and Layout Strategies on Virtual Displays

Leonardo Pavanatto, **Feiyu Lu**, Chris North, Doug A. Bowman

IEEE Transactions on Visualization and Computer Graphics (TVCG). 2024

DOI: 10.1109/TVCG.2024.3368930

CONFERENCE PROCEEDINGS

S-TIER: Situated-Traceable Insights in Extended Reality for Hybrid Crisis Management

Feiyu Lu*, Cheng Yao Wang*, Leonardo Pavanatto, Fannie Liu, David Saffo, Benjamin Lee, Mengyu Chen, Blair MacIntyre

In submission to 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2026, Bari, Italy

DOI: InSubmission

Anu.js: Accelerating Web-based Immersive Analytics

David Saffo*, Benjamin Lee*, **Feiyu Lu**, Cheng Yao Wang, Blair MacIntyre

Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems, 2026, Creant el demà junts, Barcelona

DOI: Accepted

Evaluating the Viability of Additive Models to Predict Task Completion Time for 3D Interactions in Augmented Reality

Logan Lane, Ibrahim Tahmid, **Feiyu Lu**, Doug A Bowman

2026 IEEE Virtual Reality and 3D User Interfaces (VR), 2026, Daegu, South Korea

DOI: Accepted

Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations

Matt Gottsacker, Mengyu Chen, David Saffo, **Feiyu Lu**, Benjamin Lee, Blair MacIntyre

Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, 2025, Yokohama, Japan

DOI: 10.1145/3706598.3713346

SocialMiXR: Facilitating Hybrid Social Interactions at Conferences

Fannie Liu, Cheng Yao Wang, William Moriarty, **Feiyu Lu**, Usman Mir, David Saffo, Mengyu Chen, Blair MacIntyre

Proceedings of the ACM on Human-Computer Interaction (CSCW), 2025, Bergen, Norway

DOI: 10.1145/3711069

Exploring the Impact of User and System Factors on Human-AI Interactions in Head-Worn Displays

Feiyu Lu, Yan Xu, Xuhai Xu, Brennan Jones, Laird Malamed

2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023, Sydney, NSW, Australia

DOI: 10.1109/ISMAR59233.2023.00025

In-the-Wild Experiences with an Interactive Glanceable AR System for Everyday Use

Feiyu Lu, Leonardo Pavanatto, Doug A. Bowman

Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI), 2023, Sydney, NSW, Australia

DOI: 10.1145/3607822.3614515

XAIR: A Framework of Explainable AI in Augmented Reality

Xuhai Xu, Anna Yu, Tanya R. Jonker, Kashyap Todi, **Feiyu Lu**, Xun Qian, João Marcelo Evangelista Belo, Tianyi Wang, Michelle Li, Aran Mun, Te-Yen Wu, Junxiao Shen, Ting Zhang, Narine Kokhlikyan, Fulton Wang, Paul Sorenson, Sophie Kim, Hrvoje Benko

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI), 2023, Hamburg, Germany

DOI: 10.1145/3544548.3581500

🏆 Best Paper Honorable Mention

Exploring Spatial UI Transition Mechanisms with Head-Worn Augmented Reality

Feiyu Lu, Yan Xu

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI), 2022, New Orleans, LA, USA

DOI: 10.1145/3491102.3517723

Validating the Benefits of Glanceable and Context-Aware Augmented Reality for Everyday Information Access Tasks

Shakiba Davari, **Feiyu Lu**, Doug A. Bowman

2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2022, Christchurch, New Zealand

DOI: 10.1109/VR51125.2022.00063

Exploration of Techniques for Rapid Activation of Glanceable Information in Head-Worn Augmented Reality

Feiyu Lu, Shakiba Davari, Doug A. Bowman

Proceedings of the 2021 ACM Symposium on Spatial User Interaction (SUI), 2021, Virtual Event, USA

DOI: 10.1145/3485279.3485286

🏆 Best Paper Honorable Mention

Evaluating the Potential of Glanceable AR Interfaces for Authentic Everyday Uses

Feiyu Lu, Doug A. Bowman

2021 IEEE Virtual Reality and 3D User Interfaces (VR), 2021, Lisbon, Portugal

DOI: 10.1109/VR50410.2021.00104

Glanceable AR: Evaluating Information Access Methods for Head-Worn Augmented Reality

Feiyu Lu, Shakiba Davari, Lee Lisle, Yuan Li, Doug A. Bowman

2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2020, Atlanta, GA, USA

DOI: 10.1109/VR46266.2020.00113

Gaze Direction Visualization Techniques for Collaborative Wide-Area Model-Free Augmented Reality

Yuan Li, Feiyu Lu, Wallace S Lages, Doug A. Bowman

Symposium on Spatial User Interaction (SUI), 2019, New Orleans, LA, USA

DOI: 10.1145/3357251.3357583

Evaluating Engagement Level and Analytical Support of Interactive Visualizations in Virtual Reality Environments

Feiyu Lu, Difeng Yu, Hai-Ning Liang, Wenjun Chen, Konstantinos Papangelis, Nazlena Mohamad Ali

2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2018, Munich, Germany

DOI: 10.1109/ISMAR.2018.00050

EXTENDED ABSTRACTS

Adaptive 3D UI Placement in Mixed Reality Using Deep Reinforcement Learning

Feiyu, Lu*, Mengyu Chen*, Hsiang Hsu, Pranav Deshpande, Cheng Yao Wang, Blair MacIntyre

Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, 2024, Honolulu, HI, USA

DOI: 10.1145/3613905.3651059

Clean the Ocean: An Immersive VR Experience Proposing New Modifications to Go-Go and WiM Techniques

Lee Lisle, Feiyu Lu, Shakiba Davari, Ibrahim Asadullah Tahmid, Alexander Giovannelli, Cory Llo, Leonardo Pavanatto, Lei Zhang, Luke Schlueter, Doug A. Bowman

2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2022, Christchurch, New Zealand

DOI: 10.1109/VRW55335.2022.00311

🏆 Best 3DUI Contest Entry Award

Fantastic Voyage 2021: Using Interactive VR Storytelling to Explain Targeted COVID-19 Vaccine Delivery to Antigen-presenting Cells

Lei Zhang, Feiyu Lu, Ibrahim Asadullah Tahmid, Shakiba Davari, Lee Lisle, Nicolas Gutkowski, Luke Schlueter, Doug A. Bowman

2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2021, Lisbon, Portugal

DOI: 10.1109/VRW52623.2021.00230

🏆 Best 3DUI Contest Entry Award

Achievements

2025	VR Dissertation Award , IEEE Visualization and Graphics Technical Community (VGTC)	Saint-Malo, France
2024	Prolific Inventor , Global Technology Applied Research	JPMorganChase
2023	Best Paper Honorable Mention Award (Top 5%) , CHI'23	Hamburg, Germany
2022	Best 3DUI Contest Entry Award (1 of 11 teams) , IEEEVR'22	Christchurch, New Zealand
2021	Best Paper Honorable Mention Award (Top 5%) , SUI'21	Virtual Event
2021	Finalist (Top 3.5% of 2,163 applicants) , Facebook Ph.D. Fellowship	Virtual Event
2021	Best 3DUI Contest Entry Award (1 of 17 teams) , IEEEVR'21	Lisbon, Portugal

Patents

2025	System and Method for Long-Term Video Analysis and Context-Aware Search Using Vision-Language Models , No. 19/222,406	with JPMorganChase
2024	Privacy-preserving Methods for Remote Space Sharing in Enterprise Environment , No. 18/774,705	with JPMorganChase
2024	Systems and Methods for Hybrid Cross-Reality Collaboration Around an Large Interactive Display , No. 18/882,495	with JPMorganChase
2024	HybridPortal: Enabling Hybrid Social Interactions for Hybrid Events , No. 18/882,477	with JPMorganChase
2024	Systems and Methods for Audience Feedback Guided Mixed Reality , No. 18/769,647	with JPMorganChase
2023	Systems and Methods for Placement of Virtual Content Objects in an Extended Reality Environment Based on Reinforcement Learning , No. 18/407,003	with JPMorganChase
2021	Dynamic Widget Placement Within an Artificial Reality Display , No. 17/747,767	with Meta Inc.
2020	Interacting with Glanceable Information in Wearable Augmented Reality , No. 63/147,805	with Virginia Tech

Service

2025	Program Committee - ISMAR
2025	Associate Chair (User Experience and Usability Track) - CHI
2025	Program Committee - IEEEVR
2019	Student Volunteer - IEEEVR
2018	Student Volunteer - ISMAR

Reviewing (60⁺)

2026 CHI *, IEEEVR
2025 CHI *, IEEEVR, ISMAR *
2024 CHI *, ISS *, MobileHCI, ISMAR
2023 CHI *, UIST *, ISS *, VRST
2022 IEEEVR, ISMAR *, SIGGRAPH, UIST
2021 CHI, IEEEVR, SIGGRAPH

Invited Talks

2025 **Towards Pervasive Everyday Computing**, Hosted by Dr. Hai-Ning Liang
2024 **Immersive and Intelligent XR for Enterprise Use Cases**, Hosted by Dr. Zhen Bai
2024 **Industry Panelist - Intelligent XR Workshop**, Hosted by i-XR Workshop Organizers

*The Hong Kong University of
Science and Technology
University of Rochester, NY
IEEE ISMAR, Seattle, WA*

Skills

Research	3D Interaction, Human-AI Interaction, Multimodal Input, Spatial Computing, Extended Reality, Human-Computer Interaction.
Programming/Platform	Python, C#, JavaScript/TypeScript, Swift, WebXR, VisionOS, Meta Quest.
Tools	Unity3D, BabylonJS, Arduino, Blender, R, SPSS, Adobe Suites.
Method	Rapid Prototyping, Mixed-methods Research, Inferential Statistical, Thematic Analysis, Experimental Design, Usability Evaluation, Performance Modeling.